

The Hot Magnolias — FOH EQ Rationale

Fountain Square (outdoor) · DiGiCo Quantum 225 · 2026-07-11 · Rev 3.0 Deep Think (hot-mag 3 — blind rebuild)

Artist. The Hot Magnolias — Cincinnati 8-piece New Orleans party band, formed summer 2013, CCM students/alumni plus scene vets. Repertoire runs 1920s trad-jazz standards through Mardi Gras 2nd-line to 1970s NOLA funk. Core live lineup: Jason Burkett (lead vocals), David Sharkey (sax), Chapman Sowash (trombone), Michael Cruse (trumpet), Kevin McClellan (bass), Jamarco Thomas (keys/piano), Bennett Kottler (banjo/guitar — one player doubling), Ryan Mitchell (drums). The three-horn front line and the party energy ARE the show: mix horn-forward, warm and danceable — never a scooped rock sound. Second-line snare and bass/keys pocket carry the dance floor.

Room. Fountain Square open-air: no room gain, nothing comes back from the plaza — L-Acoustics A15 mains / KS21 arch carry everything. FSQ cuts run DEEPER than indoor (–6 to –9 typical, up to –10 on mud/box); clarity is the whole game. Cuts-first everywhere, vocals cuts-only, feedback margins tight with downtown noise floor underneath. Mid-July show window: warm humid evening (~74% RH, high 86°F, scattered-shower risk) — humid air carries HF a touch better than dry, so protect presence rather than over-boosting it, and watch wind gusts on the open mics if weather moves in.

Research. Artist: CincyMusic, Bandcamp, NKyTribune. Per-source: Shure docs + PSW 91A/B52-combo thread (91A box 300–500, click up high); Gearspace/RecordingHacks D6 (pre-scooped –15 dB @ 600, needs almost no EQ); SOS/Gearspace i5 (+9 dB @ 5.5k baked peak); e609 capsule character (bright 4–5k presence, edgy 2.5–4k) from Gearspace/maker docs; DFO/Gearspace D2/D4 (D2 scooped+lean lows, D4 smooth mids/soft attack); PSW/TapeOp/Gearspace SM57-as-overheads (dark top, box 300–500, engineers open them with low-mid cuts + top lift); funk bass DI definition 700 Hz–2k (Gearspace + engineer write-ups); Banjo Hangout piezo quack 0.8–1.6k + ~5–6k brittleness; A-T Pro 35 maker/reviews (subtly rounded treble, voiced presence, thin lows, 145 dB); NOLA horn slotting — bone ~1k, sax ~2k, tpt ~3k+ (audiorecording.me, RoastYourMix); Beta 58A dual presence peaks 4k/10k (Shure spec sheet). All cross-checked against KB mic-library + eq-starting-points — no web↔KB disagreements. Show-day weather: high ~86°F, ~74% RH, scattered-shower risk (weather25/AccuWeather).

Monitors. MIX 1 Lead Vocal — Jason (wedge) · MIX 2 Horns — Sax/Tbn/Tpt front line (wedge) · MIX 3 Drums (wedge) · MIX 4 Bass + Keys (wedge) · MIX 5 Guitar/Banjo (wedge) · Counts/splits confirm at soundcheck — 8× L-Acoustics X12 on the M32.

What changed from the KB default / prior rev — and why

- Blind Rev 3 rebuild for A/B against hot-mag 2 — values derived fresh from this session's research only.
- Kick layers split explicitly: 91A owns the attack (+4 @ 5k); the D6's baked 5k click is trimmed –4 so the two mics don't stack the same peak.
- 91A contour switch assumed FLAT — the 400 Hz box cut lives on the desk. If the switch is engaged, halve the desk cut.
- D4 floor gets +2 @ 4k click — KB flags the D4's soft upper-mid attack, and open air needs the definition.
- OH 57s get +3 @ 10k as a WIDE BELL (no shelf per house rule) — justified boost on a dark dynamic outdoors with no room HF to lean on.
- Three-horn slotting: trombone owns ~1k (its box cut sits at 600), sax owns ~2k (honk cut at 1k), trumpet owns 3k+ (honk cut at 1.1k, only a –3 trim at 3.5k for PA harshness) — section separation instead of three identical horn EQs.
- Banjo quack centered at 1.2k (research consensus 0.8–1.6k) with the brittle zing cut at 5k; +2 @ 220 puts back the wood the pickup misses.
- Vocal 25/26: DEQ on the 58A's baked 4k peak + deep –8 @ 600 box cut — this OVERRIDES the FSQ template wireless curve / feedback notch on those faders. Confirm at the console.
- FSQ deeper-cut rule applied throughout: mud/box cuts sit –6 to –8.
- Hi-hat on e609 honored as specified (cab mic on a hat): HPF 500 + capsule-edge cut at 3.5k to keep it from spitting on cymbal wash.

Question round — what was asked, what Brian decided

- Rev 3 = BLIND comparison build vs hot-mag 2 (Brian, 2026-07-08): full deep build from the uploaded sheet, no values reused from Rev 2.
- Overheads: stereo pair on fader 9; fader 10 stays the SNARE PL8 return (Brian ruled, question round).
- Vocals: V1 lead (Jason) + V2 backing on Beta 58A; Vocals 3–8 stay template spares (Brian).
- No mic-locker swaps — build exactly to the sheet (Brian).
- Blank-mic channels (4, 7, 12, 16, 21–24) stay patched spares; show date 2026-07-11 confirmed.

Reverb — Seventh Heaven Pro (100% wet returns)

- **VOCAL — Plates 1 / #06 Vocal Plate** | Settings: Decay 1.5s (factory) · PreDelay 24ms · VLF –16 · E/L Max Early · Late –5 · Late Rolloff 9k · 100% wet | In-plugin EQ: LC 180 Hz on the return; keep the top bright — outdoors there's no room HF to lose | Why: The everyday lead send — dense forward plate that places Jason without wash on the up-tempo 2nd-line tunes.
- **VOCAL — Chambers 1 / #10 Vocal Chamber** | Settings: Decay 1.5s (from 1.6 factory) · PreDelay 10ms · VLF –10 · E/L Max Early · Late –6 · Late Rolloff 8k · 100% wet | In-plugin EQ: LC 160 Hz; Ducker in Reverb mode, Thr –22 dB, Release 250 ms | Why: Characterful chamber for the slower/torch numbers; the ducker clears it between phrases.
- **VOCAL — Chambers 1 / #17 Sunset Chamber** | Settings: Decay 1.9s (from 2.15 factory) · PreDelay 20ms · VLF –18 (near factory –20) · E/L Max Early · Late –6 · Late Rolloff 8k · 100% wet | In-plugin EQ: LC 180 Hz on the return | Why: The works-on-anything big-moment send — save it for the one ballad; too long for the dance material.
- **INSTRUMENT — Halls 1 / #15 Brass Hall** | Settings: Decay 1.7s (from 2.0 factory) · PreDelay 20ms · VLF –11 · E/L Max Early · Late –6 · Late Rolloff 8.4k (factory) · 100% wet | In-plugin EQ: Master Filter HS –2 @ 8k (three close clip mics feed it); LC 120 Hz | Why: One shared horn-section hall so sax/bone/trumpet read as a single front line — feature moments only; 2nd-line drive stays dry.
- **INSTRUMENT — Chambers 1 / #09 Snare Chamber** | Settings: Decay 1.0s (from 1.2 factory) · PreDelay 10ms · VLF –19 (factory) · E/L Max Early · Late –6 · Late Rolloff 5.6k (factory) · 100% wet | In-plugin EQ: LC 200 Hz on the return | Why: Short splash for the backbeat on the funk tunes — keeps the second-line snare fat without smearing the rolls.
- **GENERAL — Rooms 1 / #01 Studio A** | Settings: Decay factory · Mod 0 · E/L Max Early · Late –6 · 100% wet | In-plugin EQ: None — run the send low (3–5% equivalent), it's glue not an effect | Why: Transparent glue across kit/banjo when the plaza feels too dry — zero modulation, disappears into the sources.
- **USING THEM TOGETHER** — Using them together: Vocal Plate is the default lead send all night; switch to Vocal Chamber for the slow numbers and Sunset Chamber for the one big ballad. Brass Hall only on horn features. Snare Chamber rides the funk tunes; Studio A sits low as glue. FSQ posture per KB: near-factory decay, Late Equal to –6, bright rolloff — the plaza adds nothing.

DRUMS

Ch 1 | Kick In · Shure Beta 91A

HPF 50 | LPF off || B4 +4@5 kHz Q1.4 BELL B3 -7@400 Hz Q1.8 BELL B2 -5@200 Hz Q1.8 BELL B1 flat

Boundary on the head — beater snap plus known shell boxiness at 300–500 (KB + PSW). Contour switch would cut 400 by 7 dB; assumed flat.

+4 @ 5k gives the beater the definition that survives open air; –7 @ 400 kills the shell box deep per the FSQ rule; –5 @ 200 keeps its lows tucked so the D6 owns the body. HPF 50 for the same reason — this channel is the click engine, not the weight.

Ch 2 | Kick Out · Audix D6

HPF 30 | LPF off || B4 -4@5 kHz Q1.6 BELL B3 -4@350 Hz Q2 BELL B2 flat B1 +3@60 Hz Q1.2 BELL

Pre-scooped smiley voicing — peaks ~60 Hz and ~5k, -15 dB mid scoop at 600 baked in (Gearspace/RecordingHacks + KB agree: needs almost no EQ).

-4 @ 5k trims the D6's baked click so it doesn't stack on the 91A's +4 — the layers stay split. -4 @ 350 is a light touch on what the capsule didn't already scoop. +3 @ 60 feeds the KS21s: outdoors there's no room gain, the subs carry all the weight.

Ch 3 | Snare Top · Audix i5

HPF 100 | LPF off || B4 -3@5.5 kHz Q2.5 BELL B3 -7@500 Hz Q1.8 BELL B2 flat B1 flat

i5 bakes in +9 dB @ 5.5k presence and a +5 dB body lift ~150 (SOS/Gearspace + KB). The crack is already in the capsule.

The baked 5.5k peak IS the cut-through — just -3 @ 5.5k to take its harsh edge off the A15s. -7 @ 500 dumps the box deep per FSQ. The 150 body lift stays untouched: second-line snare wants that fatness. Ring gets handled live if the head speaks.

Ch 5 | Hi-Hat · Sennheiser e609

HPF 500 | LPF off || B4 -6@3.5 kHz Q1.8 BELL B3 -6@900 Hz Q2 BELL B2 flat B1 flat

Bright 4–5k presence voiced for guitar cabs, edgy 2.5–4k on bright sources, thin lows (helpful here). Rolled top vs a condenser — don't chase air that isn't in the capsule.

HPF 500 per the hat rule — everything below is bleed. -6 @ 3.5k tames the capsule's cab-voiced edge before it turns cymbal wash spitty; -6 @ 900 clears snare-bleed clank. In a trad-jazz set the hat rides the time — it needs to be clean, not loud.

Ch 6 | Rack 1 · Audix D2

HPF 80 | LPF off || B4 flat B3 -6@500 Hz Q1.8 BELL B2 flat B1 +2@120 Hz Q1.2 BELL

Scooped mids and voiced attack baked in; leaner deep lows than the D4 (DFO/Gearspace + KB).

-6 @ 500 finishes the cardboard the capsule's scoop started — deep per FSQ. +2 @ 120 puts back the body the D2 runs lean on. Attack is already voiced in; no top boost.

Ch 8 | Floor Tom · Audix D4

HPF 40 | LPF off || B4 +2@4 kHz Q1.4 BELL B3 -7@450 Hz Q1.8 BELL B2 flat B1 +3@80 Hz Q1.2 BELL

Reaches 35 Hz, smooth unscooped mids, less upper-mid attack than the D2 (KB flags it may need a touch of click).

The D4 doesn't scoop its own mids, so the -7 @ 450 mud cut does the work here. +3 @ 80 for weight into the subs, +2 @ 4k adds the click the capsule undersells — a funk floor tom outdoors disappears without it.

Ch 9 | OH · Shure SM57 (pair)

HPF 250 | LPF off || B4 +3@10 kHz Q1 BELL B3 -5@3 kHz Q1.8 BELL B2 -6@400 Hz Q1.6 BELL B1 flat

57s as overheads = deliberately dark, mid-forward cymbals (TapeOp/PSW consensus); box 300–500 and presence clang 3–5k are the known problems, top end rolls off early.

+3 @ 10k as a wide bell (not a shelf) opens the top the 57 doesn't have — outdoors the air dies over the throw and these are the only cymbal mics. -5 @ 3k pulls the clang the 57's presence bump makes of crash wash; -6 @ 400 dumps the kit box. HPF 250: everything below lives in the close mics.

BASS

Ch 11 | Bass DI · Whirlwind IMP (passive DI)

HPF 35 | LPF off || B4 +2@800 Hz Q1.4 BELL B3 -7@350 Hz Q1.8 BELL B2 flat B1 +3@80 Hz Q1.2 BELL

IMP is honest with a slight passive HF/level loss (KB) — no active sparkle to lean on.

+3 @ 80 is the funk punch into the KS21s; -7 @ 350 clears the mud lane deep per FSQ so keys and guitar can live there; +2 @ 800 is the definition/growl (research: 700 Hz–2k is where bass reads past the subs on a plaza). Fundamental-plus-harmonics, not just weight.

GUITAR

Ch 13 | Guitar Dyn · Shure SM57

HPF 100 | LPF off || B4 -4@3.2 kHz Q2 BELL B3 -8@400 Hz Q1.8 BELL B2 flat B1 flat

Mid-forward workhorse: presence 3–5k, box/honk builds 300–500 on cabs (KB + every forum thread ever).

NOLA funk guitar is clean chank, not high-gain wall — the 57's presence is the scratch, so only -4 @ 3.2k to keep the A15s from getting scratchy. -8 @ 400 dumps the cab box hard (FSQ deep rule); the Beta 27 fills the body cleanly instead.

Ch 14 | Guitar CON · Shure Beta 27

HPF 150 | LPF off || B4 -3@9 kHz Q2.5 BELL B3 -7@500 Hz Q1.2 BELL B2 flat B1 flat

Flat 60–3k with small peaks at 5.5k and 9k — the clean open top the 57 lacks; slight 9k fizz on bright amps (KB).

Complement rule: its low-mids come down so it can't stack on the 57's midrange — -7 @ 500 at a wide Q1.2 covers the 150–800 dump. -3 @ 9k takes the fizz off before the open air exposes it. Its job is body and top; the 57 owns the middle.

Ch 15 | Banjo · Whirlwind IMP (passive DI)

HPF 120 | LPF off || B4 -7@5 kHz Q1.6 BELL B3 -6@1.2 kHz Q1.8 BELL B2 +2@220 Hz Q1.4 BELL B1 flat

Banjo pickups carry far more HF zing than a guitar pickup plus the piezo quack; the IMP passes it all through honestly (KB: treat at the desk).

-6 @ 1.2k centers the nasal quack (Banjo Hangout consensus: 0.8–1.6k); -7 @ 5k kills the brittle zing before it ice-picks the plaza. +2 @ 220 puts back the wood/head body the pickup misses — trad-jazz banjo is a rhythm chank, it should sound like an instrument, not a transducer.

KEYS

Ch 17 | Keys · DI (XLR)

HPF 40 | LPF off || B4 -3@3 kHz Q2 BELL B3 -6@300 Hz Q1.6 BELL B2 flat B1 flat

Clean line source — EQ is mix carving, not correction (KB).

NOLA piano (the Longhair/Booker rolls) lives in the mids, so the work is making room around it: -6 @ 300 clears DI low-mid warmth against the PA, -3 @ 3k bows to the horns in their lane. HPF 40 keeps the left hand — the piano bass matters when the band drops to keys and snare.

HORNS

Ch 18 | Sax · Audio-Technica Pro 35

HPF 80 | LPF off || B4 -4@6 kHz Q1.8 BELL B3 -6@1 kHz Q1.6 BELL B2 +2@400 Hz Q1.2 BELL B1 flat

Pro 35: voiced presence lift, subtly rounded treble, thin lows, plasticky upper-mid when close (maker + KB agree). 145 dB — level is never the problem.

Sax owns the ~2k slot in the three-horn line, so its cut sits at 1k (–6, clears the bone's lane and its own honk). –4 @ 6k takes the clip-mic plastic glare; +2 @ 400 warms the thin capsule so the sax reads full, not reedy. Overtones stay — NOLA horns cut with harmonics, not volume.

Ch 19 | Trombone · Audio-Technica Pro 35

HPF 60 | LPF off || B4 -4@5 kHz Q1.8 BELL B3 -6@600 Hz Q1.8 BELL B2 +2@250 Hz Q1.2 BELL B1 flat

Same Pro 35 character — rounded treble, voiced presence, thin lows.

Bone owns the ~1k slot (research: it differentiates there), so nothing is cut at 1k — the blat/box comes out below it at –6 @ 600. –4 @ 5k for the clip glare, +2 @ 250 for the low warmth that makes a trombone sound like brass plumbing instead of a kazoo. HPF 60 keeps the pedal-range weight.

Ch 20 | Trumpet · Audio-Technica Pro 35

HPF 150 | LPF off || B4 -3@3.5 kHz Q2 BELL B3 -5@1.1 kHz Q1.8 BELL B2 flat B1 flat

Pro 35 on a trumpet bell: the voiced presence plus trumpet's natural 1–4k aggression is the thing to manage, not encourage.

Trumpet owns 3k-and-up in the slotting, so only a –3 trim at 3.5k keeps it from dominating the A15s while the brilliance above stays intact. –5 @ 1.1k clears its honk out of the bone's lane. HPF 150 per the research consensus — nothing useful below. It'll still lead the section; that's the job.

VOCALS

Ch 25 | Vocal 1 · Shure Beta 58A

HPF 120 | LPF off || B4 -4@4 kHz Q2 BELL +DEQ B3 -8@600 Hz Q1.8 BELL B2 -5@250 Hz Q1.6 BELL B1 flat

Beta 58A: two baked presence peaks (4k and 10k), sub-500 attenuated for proximity control, supercardioid = the gain-before-feedback pick outdoors (Shure spec + KB).

Cuts only. –8 @ 600 is the box cut where feedback builds first — deep per FSQ. –5 @ 250 handles the proximity mud when he eats the mic on the shouters. The 4k peak gets a DYNAMIC –4 (thr –18, 3 ms/120 ms): it only clamps when he leans in hot, so intelligibility stays the rest of the time. The 10k peak is left alone — that's the only air a cuts-only channel gets.

Ch 26 | Vocal 2 · Shure Beta 58A

HPF 120 | LPF off || B4 -4@4 kHz Q2 BELL +DEQ B3 -8@600 Hz Q1.8 BELL B2 -5@250 Hz Q1.6 BELL B1 flat

Matched to CH25.

Identical to Vocal 1 — gang vocals hit it loose and off-axis, and matching means it doubles as the lead spare without a rebuild mid-show.